CSI 5340 Assignment2

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This report represents the exploration and understanding of basic concepts, including forward propagation, back-propagation, soft-max regression, multilayer perceptron and convolutional neural network.

Answer1:

It will describe the steps and results of forward and back propagation in my manual algorithm and compare the gradients with that of automatic calculation to prove the correctness of my code. In addition, it shows loss value and the variation of loss based on my algorithm while using one X and N of X (Figure 3).

There are 5 layers: Layer1 (y=Ax), Layer2 (u=σ (y)), Layer3 (v=Bx), Layer4(z=u + v), Layer5(w=Cz). L1, L3, L5 are multiplication layers, L2, L4 are sigmoid layer and addition layer respectively, and all of layers include forward and backward functions which will use function called mul( ) in my code to achieve multiplication.

Forward propagation: using forward function of Layer1, Layer2, Layer3, Layer4, Layer5 one by one, and calculating loss (L).

Brief description of back-propagation:

Layer5.backward( ) : input: d_w (dl/dw)

output: d_C (dw/dc), d_z (dw/dz)

Layer4.backward( ) : input: d_z

output: d_u (dz/du), d_v (dz/dv)

Layer3.backward( ) : input: d_v

output: d_B (dv/dB), null

Layer2.backward( ) : input: d_u

output: d_y(sigmoid`(y))

Layer1.backward( ) : input: d_y

output: d_A (dy/dA), null

d_w = dL/dw = w[K]/L

d_C = dL/dw · dw/dC = d_w · z

d_z = dL/dw · dw/dz = d_w · C

d_v = dL/dw · dw/dz · dz/dv = d_z ·1 = d_z

d_u = dL/dw · dw/dz · dz/du = d_z ·1 = d_z

d_B = dL/dw · dw/dz · dz/dv · dv/dB = d_v · x

d_y = sigmoid`(y)

d_A= dL/dw · dw/dz · dz/du · du/dy · dy/dA = d_y · x

The above-mentioned equations are useful to implement the back propagation. All the results are showed below, which prove that the derived network is correct (comparing Figure1 with Figure2)

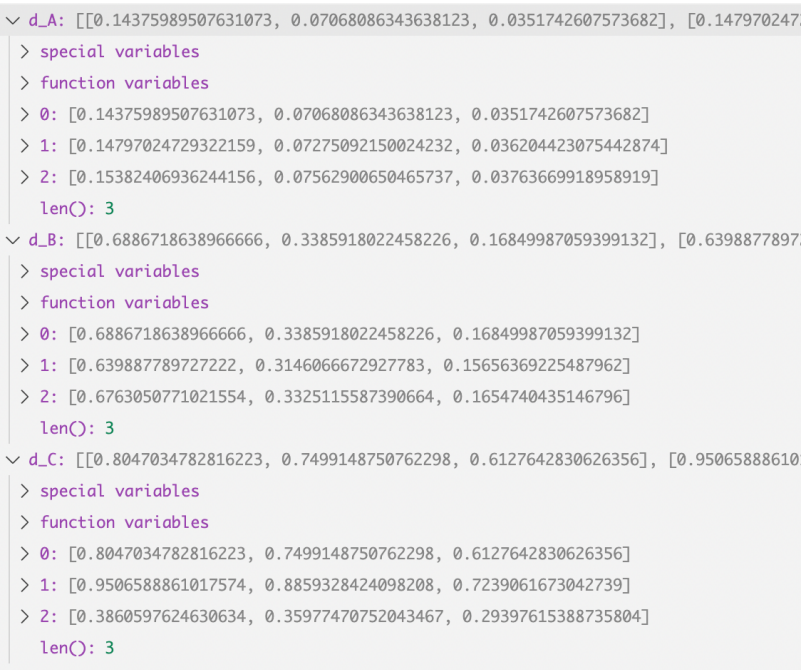


Figure 1: execution result of my program

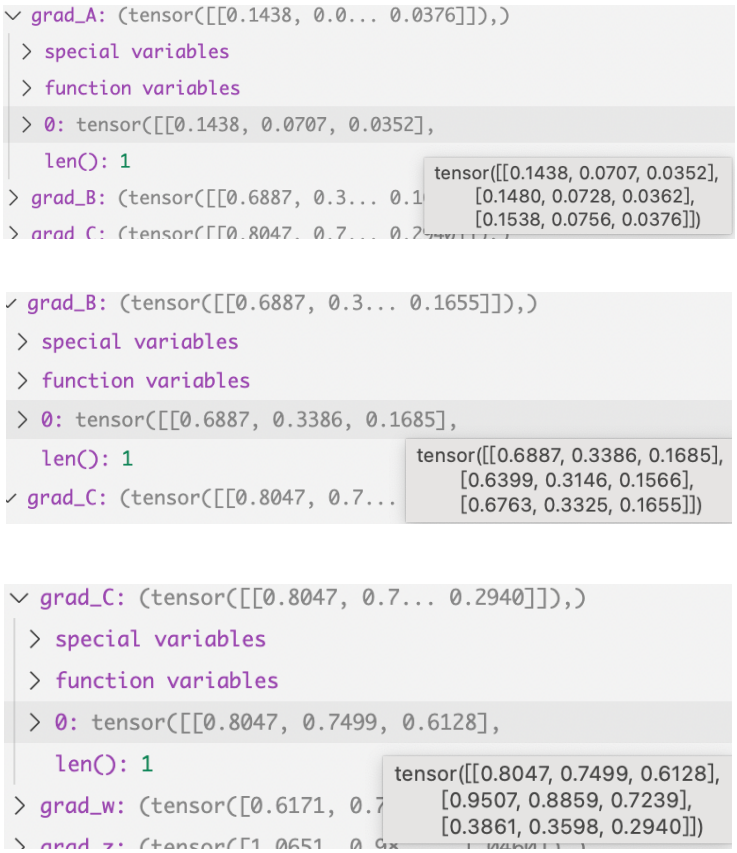


Figure 2: results using torch

In Figure3, the left one shows the variation in MSE after 100 epoch when the number of X is one, the right present MSE trend with N of X (N=15) after 150 epoch.

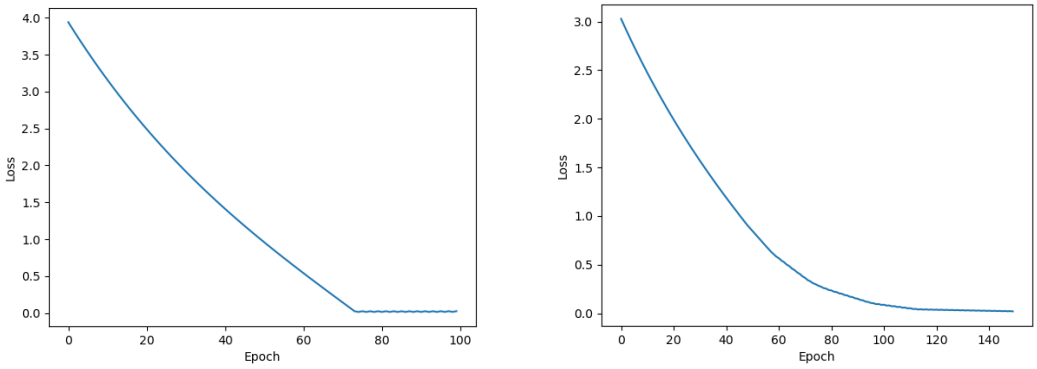


Figure 3 MSE

Answer2:

First, showing and comparing the variation in the loss and accuracy of three classifiers when use 6 types of structure, including:

without Batch Normalization, without Dropout (original model)

without Batch Normalization, with Dropout (Dropout rate=0.8)

without Batch Normalization, with Dropout (Dropout rate=0.5)

without Batch Normalization, with Dropout(Dropout rate =0.2)

with Batch Normalization, without Dropout

with Batch Normalization, with Dropout.

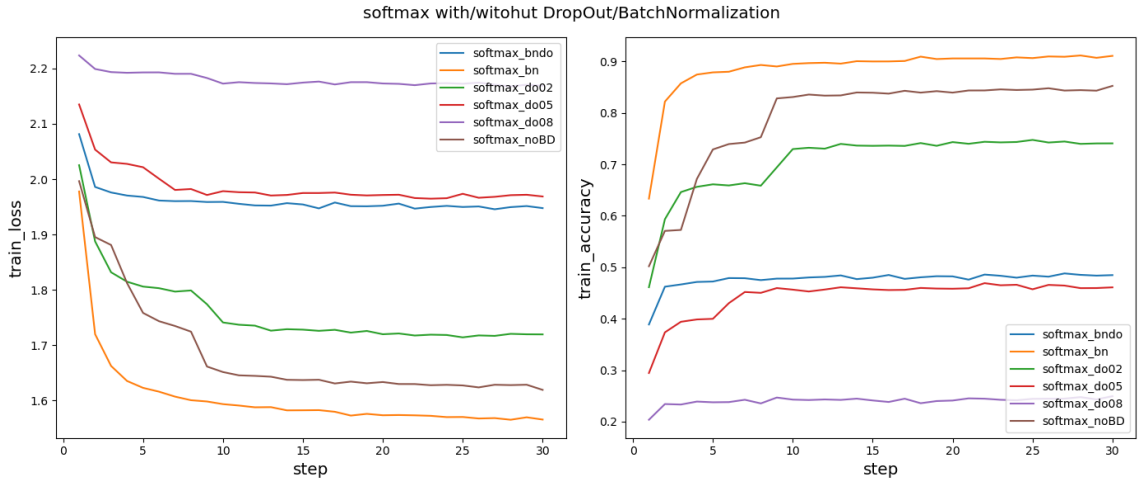
Then, presenting the best-case differences of three classifiers. All of them are using Batch Normalization and same parameters.

The influence of Batch Normalization and Dropout

- Softmax:

  Loss function: cross entropy

  Optimizer: SGD



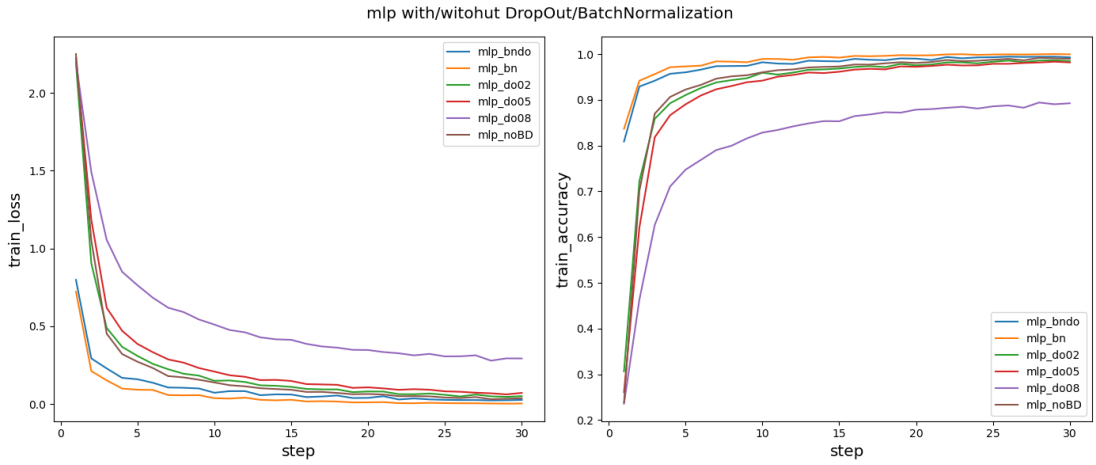
Both Batch Normalization and Dropout can converge the model even faster. To be noticed, the former increases the accuracy significantly, contributing to the best performance with above 90% accuracy and lower than 1.6 loss, while the latter damages a lot resulting in lower accuracy and higher loss compared with original model, especially 2.18 loss and 35% accuracy appears in Dropout=0.8.

In addition, applying Batch Normalization and Dropout at the same time decreases the performance of model with Soft-max Regression. (1+1<2)

- MLP:

  Layer1, Layer2, Layer3, Layer4: 512nodes, 256 nodes, 128nodes, 64nodes

  Activation function: Relu



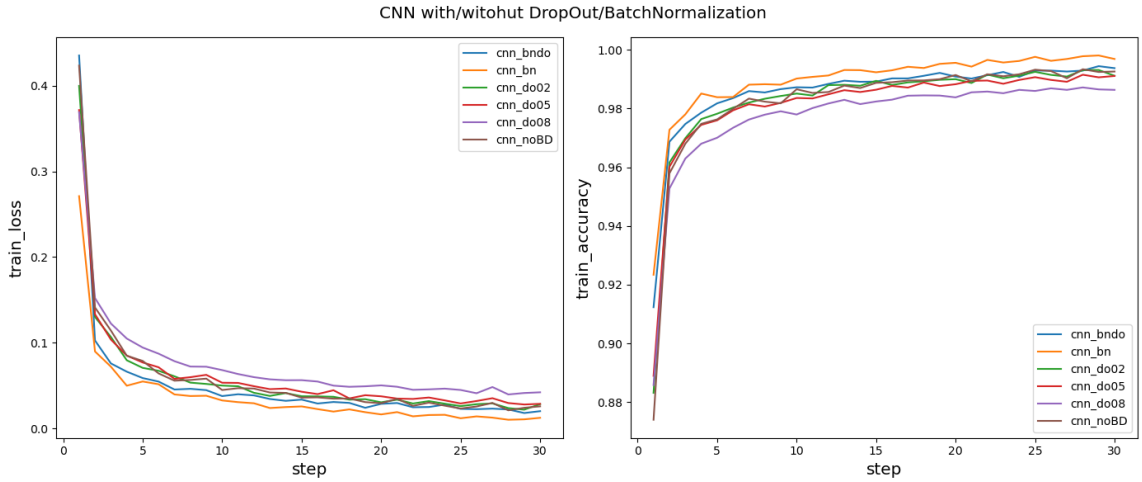
Apparently, increased accuracy, declined loss and faster convergency occur in MLP with Batch Normalization to 99%, 0.002, 5epoch respectively. Although MLP with both BN and Dropout generates similar effect, it does not surpass the only normalized one.

MLP with Dropout 0.2, which shows a similar picture with original model (neither Dropout nor BN), is a little better than that with Dropout 0.5. Strikingly, Dropout 0.8 still leads to the worst performance among all such as lower accuracy, higher loss and slower convergency.

- CNN:

  2 layers convolutional neural network

  Activation function: Relu



The performance of CNN model with Batch Normalization is the best, which presents the lowest loss (0.01) and the highest accuracy(99.6%). Instead, Dropout=0.8 still performs the worst, and even is exceeded by original model (CNN_noBD). Moreover, CNN with Dropout 0.2, with BN+Dropout and original model are the same in overall tendency of loss, accuracy and convergency, and the performance of mentioned are better than that with Dropout 0.5, but it is far less important.

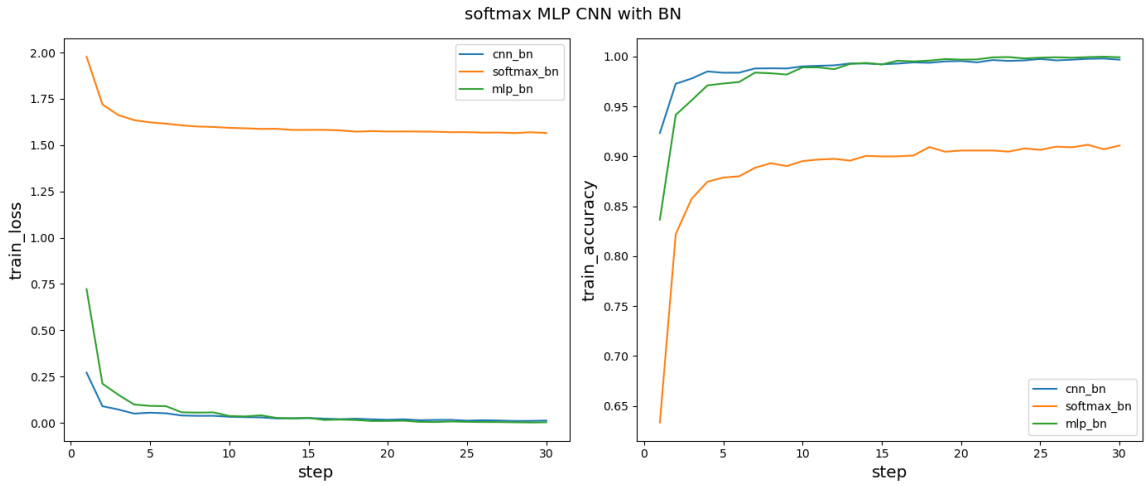
- Conclude:

Both Batch Normalization and Dropout can accelerate convergency, but the former exerts the most ideal results on the loss and accuracy than the latter even than all the other methodologies.

In addition, the differences caused by different dropout rate are huge. To be specific, the higher dropout rate is, the worse results will be generated. Dropout 0.8 always damages performance in every cases. Surprisingly, even the optimized effect of dropout 0.2 is not obvious, which causes similar loss and accuracy with original model in most situations.

Furthermore, it is not necessary to use BN and Dropout at the same time, since the influence is 1+1<2 which is not better than BN or even original model.

Comparison of Soft-max regression, MLP, CNN



Parameter setting:

Optimizer: SGD

Dropout rate: 0.5

Learning rate: 0.01

Batch size: 300

Epoch: 10

Apparently, when parameters are the same, CNN and MLP draw similar pictures, whereas soft-max regression presents the poorest performance.